

9.8 SCORING ELEMENTS

SCORING ELEMENTS for BIOBUZZ are POLLEN and NECTAR.

- POLLEN are approximately 2.8 in. (7.1 cm) Gopher ResisDent™ polyethylene balls in yellow ([am-5851_yellow](#)).
- NECTAR are approximately 3.6 in. (9.1 cm) Gopher ResisDent™ polyethylene balls in red ([am-5852_red](#)) and blue ([am-5852_blue](#)).

There are 40 POLLEN, 8 red NECTAR, and 8 blue NECTAR total in a BIOBUZZ MATCH.

POLLEN and NECTAR are not perfectly spherical and may vary in size. Teams should plan for this variation when designing their ROBOTS.

Figure 9-13: SCORING ELEMENTS: POLLEN



Figure 9-14: SCORING ELEMENTS: NECTAR

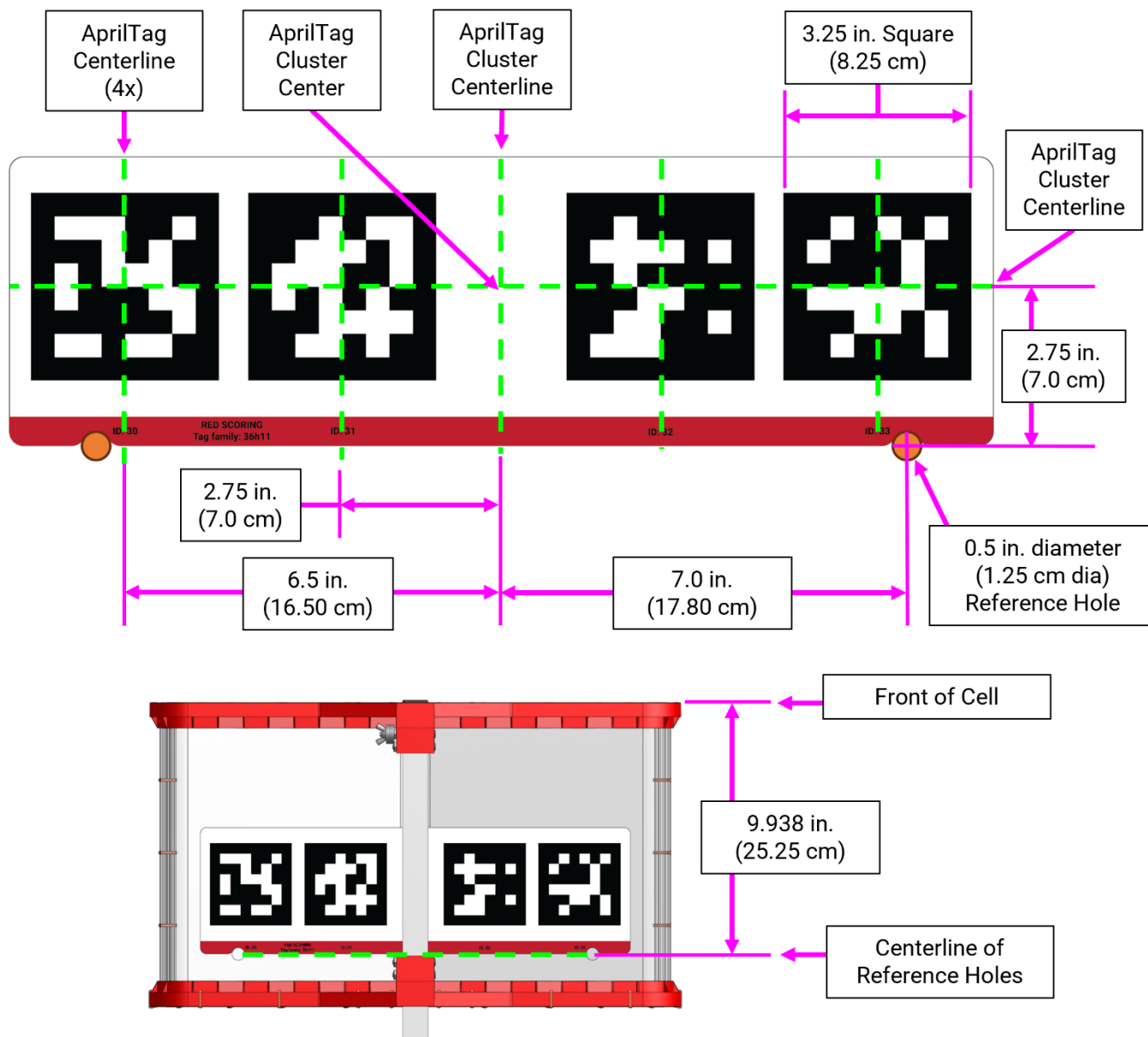


9.9 AprilTags

AprilTags for BIOBUZZ are 3.25 in. (8.25 cm) square targets from the 36h11 tag family (Figure 9-17) which are used to aid in ROBOT navigation and targeting.

AprilTags are arranged in AprilTag Clusters, consisting of four AprilTags laid out on a single sticker. Each individual AprilTag has an identifying “ID” text label. The AprilTag Clusters are applied to the CELLS as part of a single sticker, that is aligned using Reference Holes. These Reference Holes can be used to measure the location of the AprilTag Cluster relative to the rest of the FIELD. AprilTag dimensions are shown in Figure 9-15.

Figure 9-15: AprilTag Cluster and Reference Hole layout

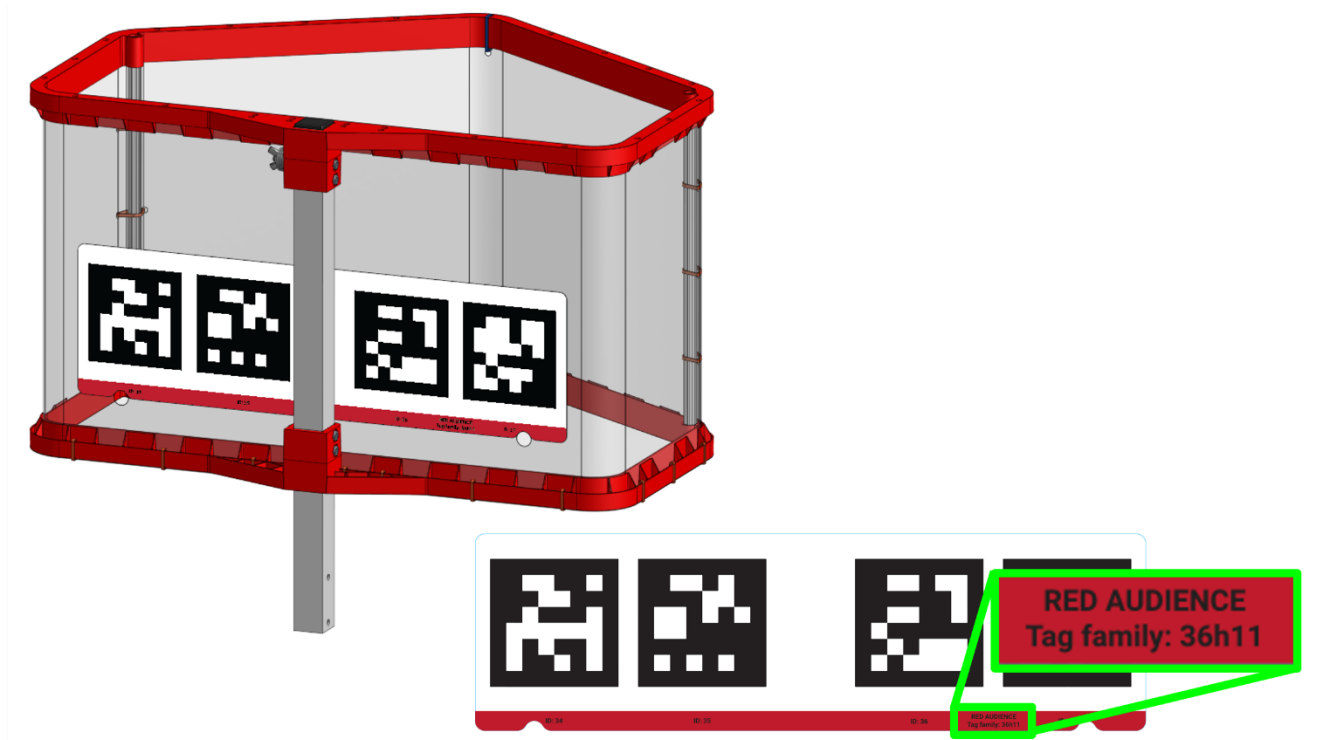


Images from this manual are examples and not to scale and not intended to be printed for practice purposes.

Please refer to the [Playing Field Resources page](#) for printable versions of these images.

Each AprilTag Cluster is placed on the bottom of a CELL facing downward towards the TILES with its bottom edge oriented towards the center of the FIELD (Figure 9-16).

Figure 9-16: AprilTag Cluster on CELL



The AprilTags are located in the following locations on the FIELD, and in the order indicated in Figure 9-17:

- AprilTag ID's 30, 31, 32, 33 on the red CELL on the side of the FIELD opposite of the audience.
- AprilTag ID's 34, 35, 36, 37 on the red CELL on the audience side.
- AprilTag ID's 38, 39, 40, 41 on the blue CELL on the audience side.
- AprilTag ID's 42, 43, 44, 45 on the blue CELL on the side of the FIELD opposite of the audience.

Figure 9-17: AprilTag Locations on the BIOBUZZ FIELD

